



Scapy: The Packet Manipulation Library

The third article in the ongoing series on FOSS security tools introduces the Scapy packet program and library.

Scapy is a packet manipulation library written in Python that can also be used interactively. You can create and decode packets for a number of protocols, and also store them in a packet capture (pcap) file. It can perform scans, probes, network discovery, traceroutes, attacks, and has many features of tools like Nmap, tcpdump, Wireshark, etc. Released under the GNU GPL 2.0 licence, its source code is available on GitHub. It requires at least Python 3.7 and has cross-platform support.

Installation

You can install Scapy on Ubuntu using the following command:

```
$ sudo apt install python3-scapy python3-pyx python3-geoip python3-cartopy
```

The pyx package is useful for generating PostScript graphics output. The python3-geoip and python3-cartopy packages can produce traceroute and visualisation maps. The version of Python3 used for this article is 3.10.12 from Ubuntu 22.04.4 LTS system as shown below:

```
$ python3 --version
Python 3.10.12
```

You can invoke the 'scapy' tool directly from the command line, but you will need to use it with the 'sudo' command for privileged commands.

```
$ sudo scapy
...
```

```

aSPY//YASa
apyyyyCY////////YCa      |
sY////////YSpcs  scpCY//Pp | Welcome to Scapy
ayp ayyyyyySCP//Pp      syY//C | Version 2.4.4
AYASAYYYYYYYY//Ps      cY//S  |
pCCCCY//p      cSSps y//Y  | https://github.com/
secdev/scapy
SPPPP///a      pP///AC//Y  |
A//A      cyP///C  | Have fun!
p///Ac      sC///a  |
P///YCpc      A//A  | Craft packets like I
craft my beer.
scccccp///pSP///p      p//Y  |
Jean De Clerck
sY/////////y  caa      S//P  |
cayCyayP//Ya      pY/Ya
sY/PsY////YCcc      aC//Yp
sc  sccaCY//PCypaapyCP//YSs
spCPY////////YPSps
ccaacs
using IPython 7.31.1

>>>
```

Configuration

The 'lsc()' command displays the default functions in Scapy: